	School of Engineering & Technology	
	Department: SOET	Session: 2026-2027
	Program: B.Tech/ BCA/ BSc	Semester:3rd
	Course Code: ETCCDA302/ ETCCDA301/ ETCCAD301/	Number of students:
	Course Name: Design and Analysis of Algorithms	Faculty: Ms. Ritu Devi

Question Bank: Unit-1

Short Answer Type Questions

1. Explain the properties of an algorithm with an example.

2. Distinguish between Algorithm and Pseudocode

3. List out the steps that need to design an algorithm.

4. Explain the key characteristics of a good algorithm

5. Define time complexity and space complexity.

6. What is the significance of the best-case time complexity? Can you give an example of an algorithm with different best-case and worst-case time complexities?

7. What is the significance of analyzing the space complexity of an algorithm? Explain the components of space complexity. Provide an example of an algorithm where space complexity is crucial.

8. Explain the components of time complexity.

9. Differentiate between BigOh and omega notation with example

10. Compute the complexity of the following algorithm using step count

```
for i ← 1 to n do
    for j ← 1 to n do
        if i ≠ j then
            swap(A[i], A[j])
        end if
    end for
end for
```

11. What is step count method? Compute the complexity of the following algorithm using step count

```
Algorithm Mean(A, n)
    sum ← 0
    for i ← 1 to n do
        sum ← sum + A[i]
    end for
    return sum / n
end Mean
```

Long Answer Type Questions

1. Write the asymptotic notations used for best case, average case and worst-case analysis of algorithms and write an algorithm for finding maximum element of an array perform best, worst and average case complexity with appropriate order notations.
2. Solve the recurrence relation T(n)=2T(n/2)+O(n) using the Recursion Tree method.
3. Explain the insertion sort algorithm. Explain it with the help of an example. Derive the time complexity of merge sort and express it using Big-O notation.
4. Consider $T(n) = T(n/3) + T(2n/3) + cn$. Solve using a recursion tree.
5. Solve the following questions using Master Theorem:
   i. $T(n) = 3T(n/2) + n^2$
   ii. $T(n) = 4T(n/2) + n^2$
6. Solve the following questions using Master Theorem:
   i. $T(n) = T(n/2) + 2^n$
   ii. $T(n) = 2T(n/2) + n \log n$
7. Solve the following recurrence using iteration method:
   $$T(n) = 2T(n/2) + n$$
8. Consider the following recursion algorithm

```
1  Algorithm Process(n)
2      if n ≤ 1 then
3          return
4      end if
5      Process(⌊n/2⌋)
6      Process(⌊n/2⌋)
7      for i ← 1 to n do
8          print("*")
9      end for
10 end Process
```

   a) What does this algorithm compute?
   b) Setup a recurrence relation for the algorithms basic operation count and solve it.
9. Consider $T(n) = T(n/3) + T(2n/3) + cn$. Solve using iteration method.
10. Find the time complexities of the following algorithms:

(a)

```
k ← 0
i ← 1
while i ≤ n do
    j ← 2
    while j ≤ n do
        k ← k + n/2
        j ← j × 2
    end while
    i ← i + 1
end while
```

(b)

```
Algorithm Fun(n)
    if n ≤ 1 then
        return
    end if
    print(n)
    Fun(⌊n/2⌋)
end Fun
```

(c)

```
i ← 1
while i < n do
    for j ← 0 to i - 1 do
        print(i, j)
    end for
    i ← i × 2
end while
```

(d)

```
k ← 0
i ← 0
while i ≤ n do
    j ← 2
    while j ≤ n do
        k ← k + n/2
        j ← j × 2
    end while
    i ← i + 1
end while
```

11. For each of the following recurrences, give an expression for the runtime T (n) if the recurrence can be solved with the Master Theorem. Otherwise, indicate that the Master Theorem does not apply.

    a) $T(n) = 3T(n/2) + n^2$, $T(0)=1$
    b) $T(n) = 2T(n/2) + n \log n$, $T(0)=1$